Flood Hazard, Vulnerability and Risk in Lokoja, Nigeria

Validation-supported GIS Multi-Criteria Decision Analysis

Abdullah Abdazeez Ayomide

Project overview

This project identifies areas of comparatively high physical flood susceptibility in Lokoja LGA and examines how the pattern changes when population vulnerability is introduced. The hazard model combines eight flood-conditioning factors using AHP/MCDA and is tested against satellite-derived flood evidence from 2012, 2018 and 2022.

Planning question

Where is physical flood susceptibility concentrated in Lokoja, how well does the model correspond with past flood evidence, and how does human vulnerability change the final risk pattern?

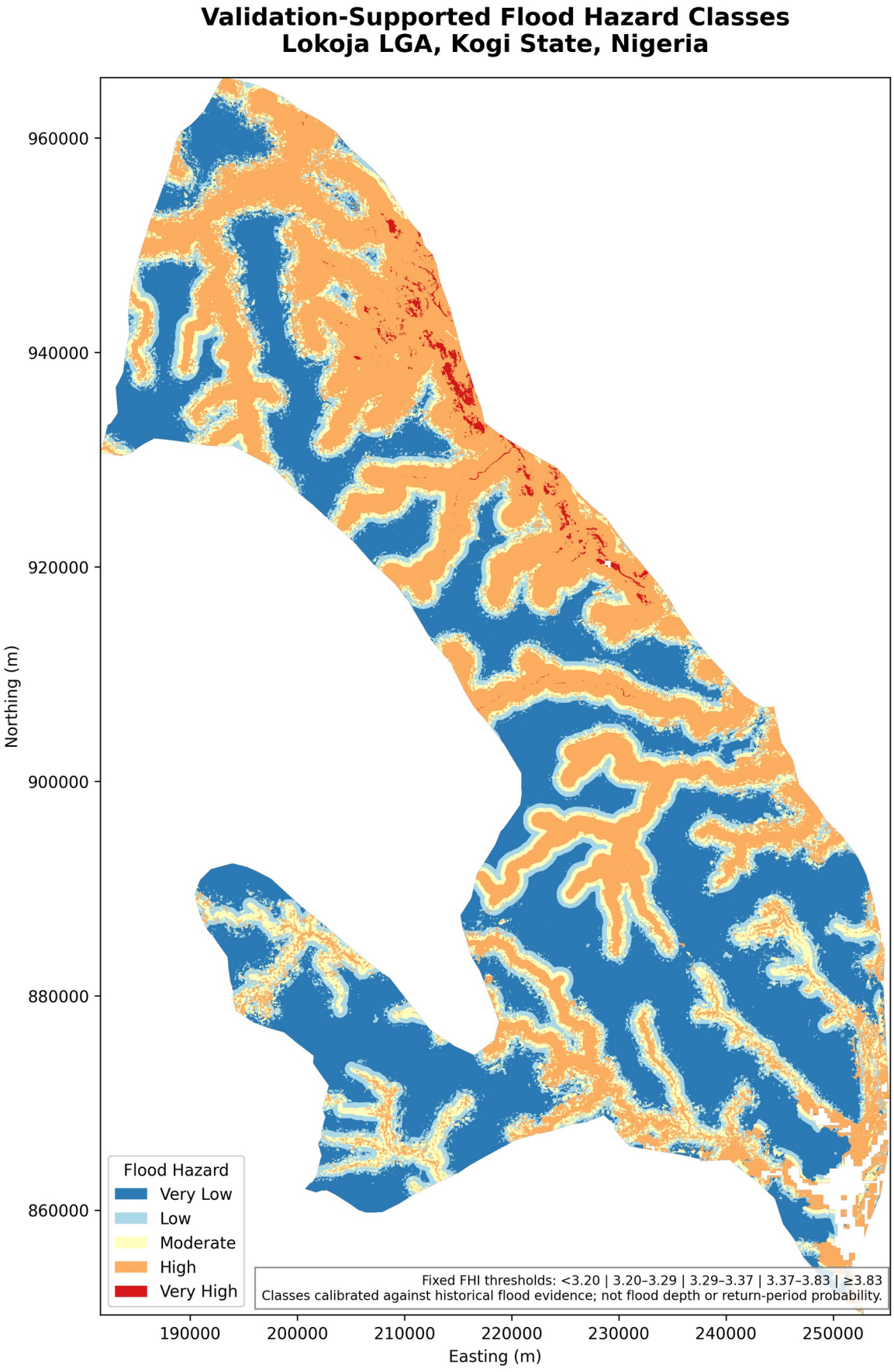


Figure 1. Validation-supported physical flood-hazard map.

Method

Distance to drainage, elevation, rainfall, slope, drainage density, land cover, clay content and NDVI were standardised and combined using AHP weights. The documented consistency ratio is 0.0093. Final hazard classes use fixed FHI thresholds of 3.20, 3.29, 3.37 and 3.83.

Indicator	Result
Distance to drainage	24.58%
Elevation	20.29%
Rainfall	15.85%
Slope	12.18%
Drainage density	9.79%
Land cover	7.37%
Clay	6.23%
NDVI	3.71%

Key results

Indicator	Result
High + Very High hazard	1,136.71 km²
Very High hazard	22.22 km²
2012 historical capture	85.08%
2018 historical capture	86.64%
2022 historical capture	86.36%
Mean AHP class agreement	91.21%
Minimum FHI rank correlation	0.99666
DEM -10 m / +10 m agreement	85.43% / 85.48%

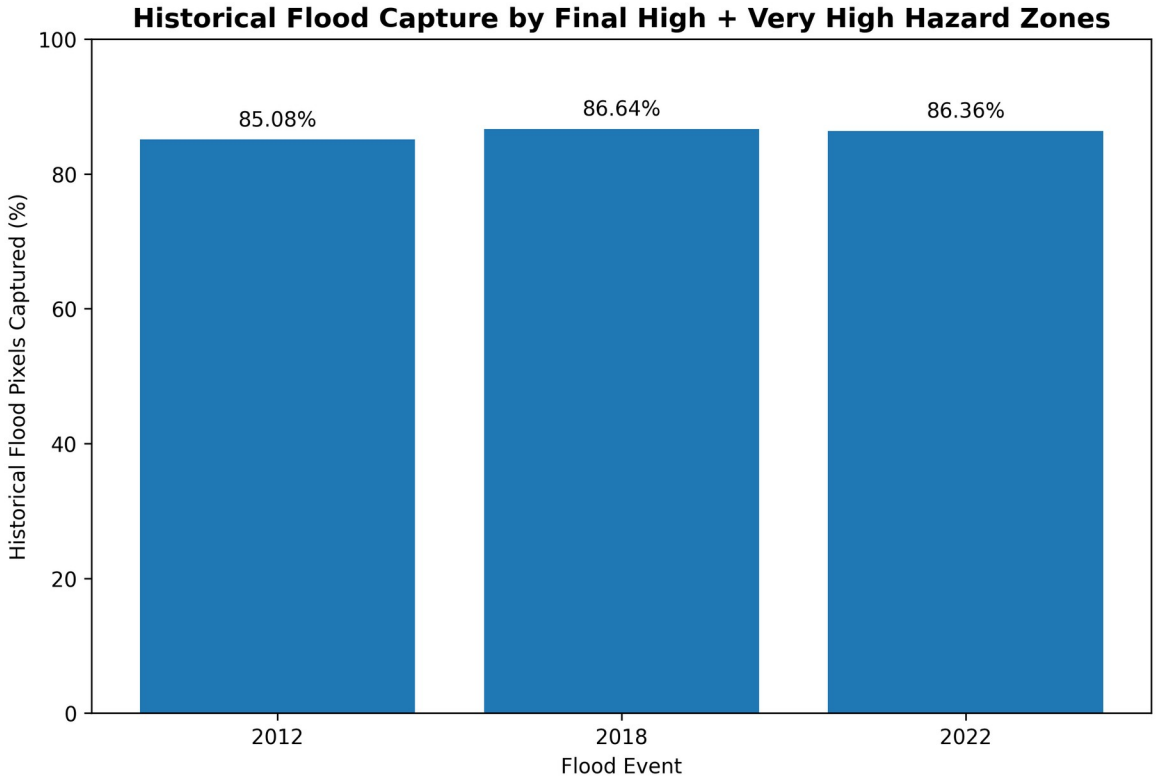


Figure 2. Historical flood evidence captured by the final hazard classes.

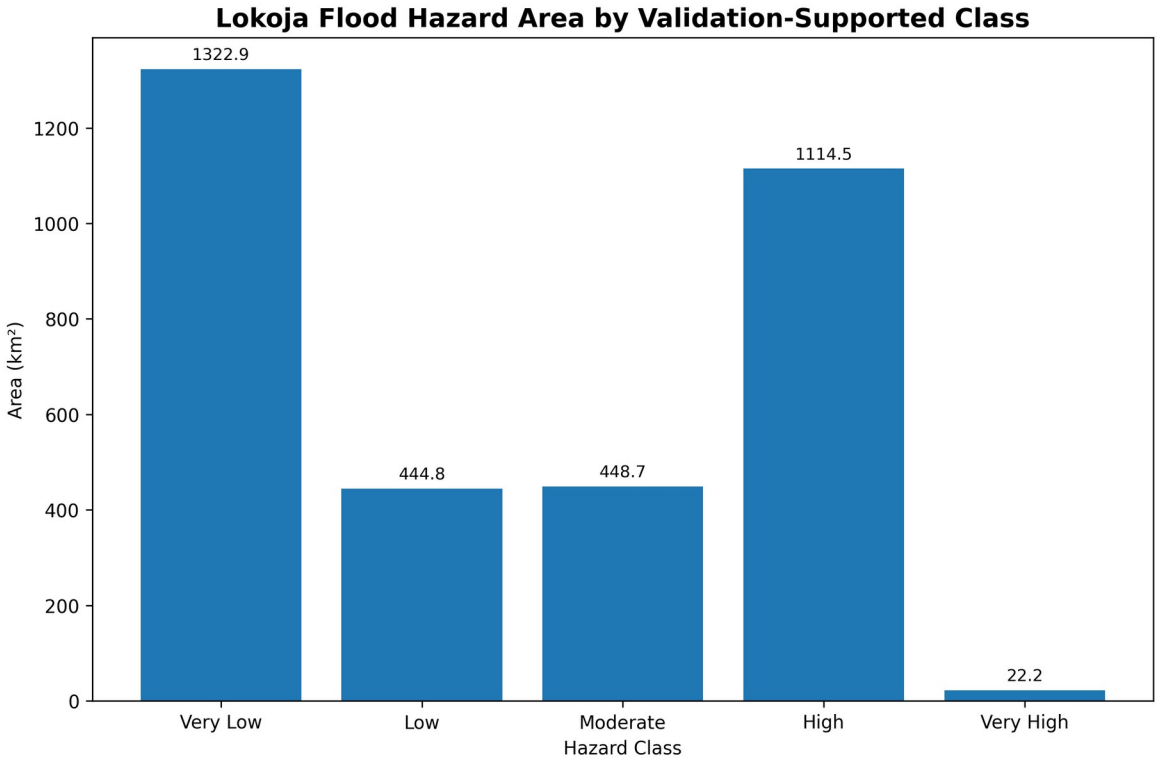


Figure 3. Area distribution of the final hazard classes.

Hazard and population-informed risk

The two products answer different questions. Physical hazard represents relative flood susceptibility. Population-informed risk introduces human vulnerability/exposure. Within the common analysis domain, 1,125.64 km² is High or Very High physical hazard, while 58.54 km² remains High or Very High in the final risk classification.

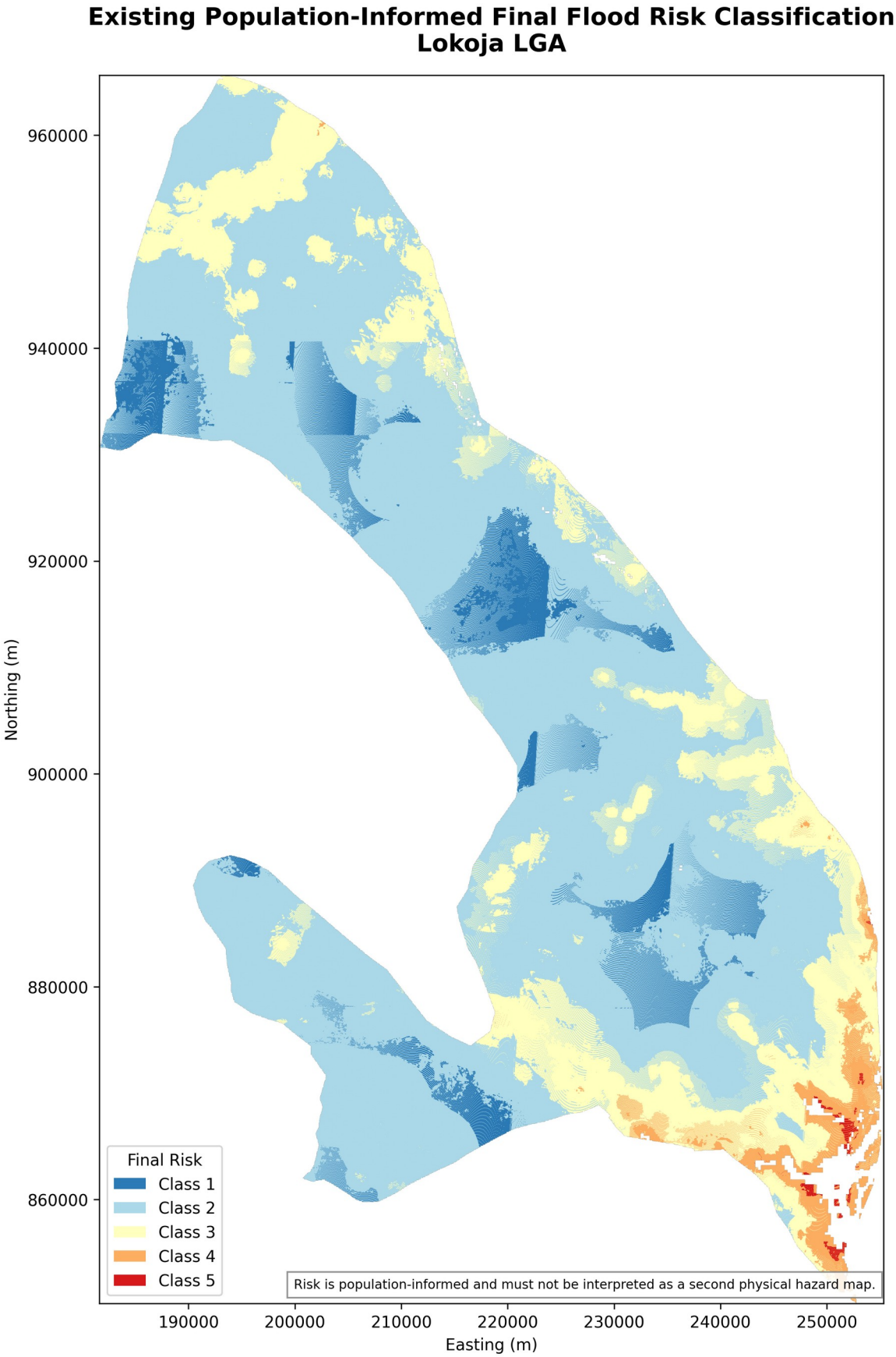


Figure 4. Population-informed final risk interpretation.

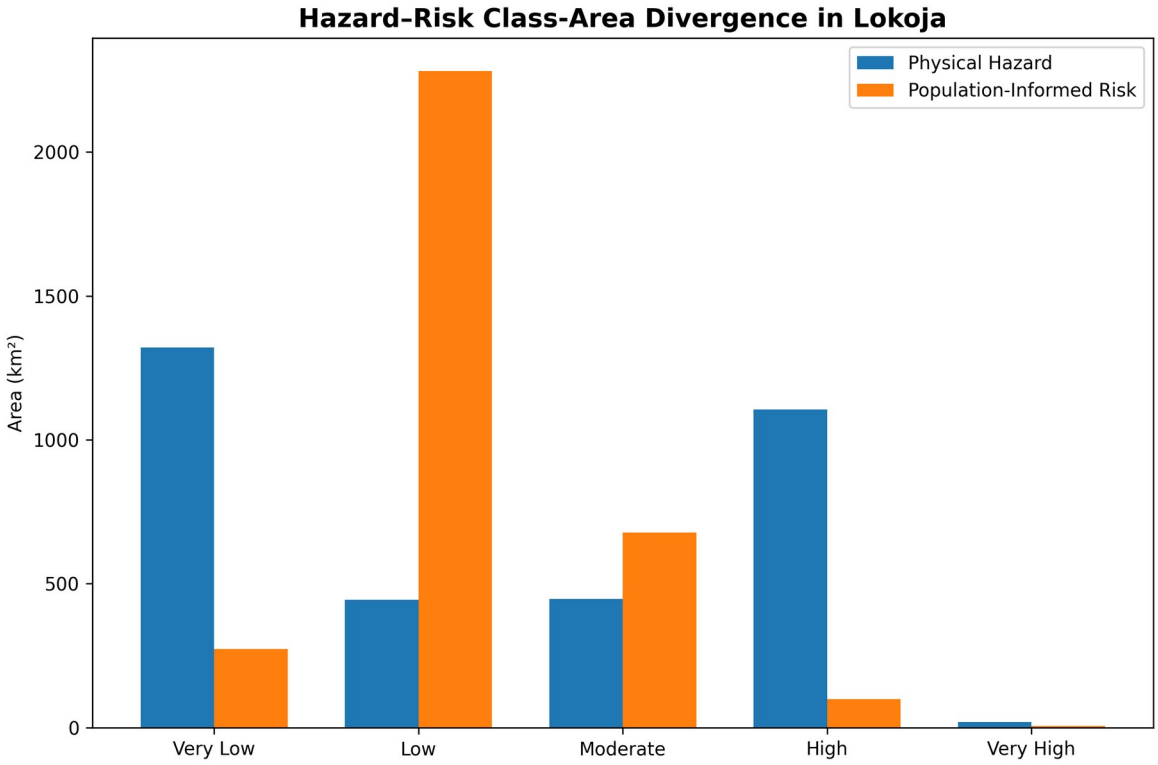


Figure 5. Difference between physical hazard and population-informed risk class areas.

Planning value

- Use physical hazard as an early screening layer for development control and drainage planning.
- Prioritise locations where high susceptibility overlaps with settlements and critical infrastructure for more detailed investigation.
- Use the population-informed risk layer for exposure-focused planning rather than as a second hazard map.
- Update the assessment when newer terrain, rainfall, land-cover, population and flood-observation data become available.

Scientific limitations

This assessment maps relative flood susceptibility. It does not estimate flood depth, annual exceedance probability, return period or hydraulic flow behaviour. The DEM sensitivity tests also show that categorical classes can change under plausible elevation offsets in Lokoja's low-relief environment. The outputs should therefore complement, not replace, hydraulic modelling and site-level engineering analysis.

Tools

Google Earth Engine · Python · Rasterio · GIS · Remote Sensing · AHP/MCDA · Historical Flood Validation

Citation

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